

A COMPREHENSIVE RECONSTRUCTION OF THE IUT-*abc* PROOF PROGRAMME PROVEN COMPONENTS, COUNTERMODELS, NORMALIZATION CORRECTIONS, NON-CIRCULAR CLOSURE, AND THE EXACT REMAINING CONSTRUCTION THEOREMS

CHATGPT

ABSTRACT. This paper consolidates the mathematical and formal results obtained in a long-form reconstruction of the proposed inter-universal Teichmüller (IUT) route to the *abc* conjecture. The work separates three layers that must not be conflated: elementary and categorical implications that can be proved independently of the full IUT source; source-level reconstructions of the pilot bridge; and the actual inhabitation theorems required to turn the public Lean specification into a parameter-free proof.

The following components are proved in ordinary mathematics and encoded in Lean source: representative extraction from equality of composite poly-morphisms; Kummer conjugation for full poly-isomorphisms; a finite countermodel showing that projection-wise compatibility is insufficient; generation of a theta-pilot region as the union of concrete outputs; finite-positive log-volume monotonicity; ramification-correct q-pilot normalization; product-weight marginalization; procession averaging; finite-exception absorption; the scalar IUT IV algebra; the geometric tripod to logarithmic *abc* transfer; the correct pointwise quantifier structure; and a circularity theorem showing that a former downstream certificate already contained the *abc* conjecture as a field.

A non-circular final certificate is defined. Its inhabitation would produce a parameter-free Lean theorem stating the standard logarithmic *abc* conjecture. No inhabitant is constructed here. The remaining obligations are an actual initial-theta-data family, an actual IUT III Hodge-theater/Frobenioid/multiradial source family, and a uniform IUT IV geometric height bridge. The public IUT formalization deliberately leaves these objects behind interfaces. Consequently this paper is a comprehensive reconstruction and formalization audit, not an unconditional proof of *abc*.

Final status

The paper proves a large collection of genuine lemmas and an honest non-circular closure theorem

$$\text{FourStageProgram} \implies \text{ABCConjecture}.$$

It does not prove

$$\text{Nonempty}(\text{FourStageProgram}).$$

Thus no theorem of the form

$$\text{theorem abc_conjecture} : \text{ABCConjecture}$$

has been obtained without parameters.

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1. THE TARGET STATEMENT

For a positive integer n , let

$$\text{rad}(n) = \prod_{p|n} p$$

be the product of the distinct prime divisors. The logarithmic form of the *abc* conjecture is the following.

Theorem 1.1 (ABC conjecture — target statement). *For every $\varepsilon > 0$ there exists $C_\varepsilon \in \mathbb{R}$ such that, for all positive pairwise coprime integers a, b, c with $a + b = c$,*

$$\log \max\{a, b, c\} \leq (1 + \varepsilon) \log \text{rad}(abc) + C_\varepsilon.$$

The formal development defines exactly this proposition. The issue is not the syntax of the final statement. The issue is constructing, uniformly in the arithmetic input and in ε , all source and height-theoretic data needed to prove it.

2. THE IUT NUMERICAL CORRIDOR

The decisive IUT III corridor is usually displayed as

$$W \in \mathcal{U}_\Theta, \quad \mu_{\text{proc}}(W) = -|\log q| \rightarrow -|\log q| \leq -|\log \Theta| \rightarrow C_\Theta \geq -1.$$

The first implication uses monotonicity of the procession-normalized volume; the second combines the bridge with

$$-|\log \Theta| \leq C_\Theta |\log q|, \quad |\log q| > 0.$$

The downstream IUT IV argument is intended to turn $C_\Theta \geq -1$ into a Vojta-type inequality on the tripod $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and then into [Theorem 1.1](#).

The research programme therefore decomposes into four stages:

- (1) construct an actual family of initial theta-data;
- (2) construct an actual IUT III multiradial output family and the witness W ;
- (3) prove a uniform, non-circular IUT IV geometric bridge;
- (4) compose the preceding stages to obtain the parameter-free *abc* theorem.

3. WEAK COMPATIBILITY IS NOT ENOUGH

The first important negative result is that a common source, a horizontal pilot map, compatibility on projected unit data, and the presence of an ordinary Kummer branch do not by themselves force a coherent pilot output.

Counterexample 3.1 (Finite common-source countermodel). *Let*

$$A_0 = A_1 = C_0 = C_1 = \{0, 1\}.$$

Take the theta-pilot to be 0, the q-pilot to be 1, and let the horizontal map interchange 0 and 1. Let both Kummer maps be identities. Suppose the only allowed coric transport is the identity. Make every unit and label projection a singleton, let Ind2 act trivially, and include an ordinary branch whose possible output is 0.

Then all projected compatibility conditions hold and the horizontal map sends the theta-pilot to the q-pilot, but no allowed coric map sends the Kummer image of 0 to the Kummer image of 1.

This counterexample does not refute IUT. It refutes the inference from a list of weak projection-wise properties to the existence of one object-level coherent tuple. It also explains why a formalization must identify the exact level at which fullness or compatibility is asserted.

4. POLY-MORPHISM REPRESENTATIVE EXTRACTION

Let

$$K_0 : A_0 \xrightarrow{\sim} C_0, \quad K_1 : A_1 \xrightarrow{\sim} C_1$$

be Kummer isomorphisms. Let

$$H \subseteq \text{Iso}(A_0, A_1), \quad P \subseteq \text{Iso}(C_0, C_1)$$

be poly-isomorphisms. Define

$$K_1 H = \{K_1 \circ h : h \in H\}, \quad PK_0 = \{p \circ K_0 : p \in P\}.$$

Theorem 4.1 (Representative extraction). *If*

$$K_1 H = PK_0,$$

then for every $h \in H$ there exists $p \in P$ such that

$$pK_0 = K_1 h.$$

Proof. The composite $K_1 h$ lies in $K_1 H$. By equality of the two composite sets, it lies in PK_0 , hence equals pK_0 for some $p \in P$. $\square$

This elementary theorem was one of the first points to be formalized in Lean. It isolates the precise logical difference between compatibility of complete composite poly-morphisms and separate compatibility of their projections.

5. FULL-POLY CONJUGATION

If the coric relation is the full poly-isomorphism, the explicit equality of composite sets is unnecessary.

Definition 5.1 (Kummer conjugate). Given a horizontal isomorphism $h : A_0 \simeq A_1$, define

$$P = K_1 h K_0^{-1} : C_0 \simeq C_1.$$

Theorem 5.2 (Full-poly conjugation). *For every $x \in A_0$,*

$$P(K_0 x) = K_1(hx).$$

If the allowed coric relation is the full poly-isomorphism $\text{Iso}(C_0, C_1)$, then P is automatically an allowed representative.

Proof. The first assertion is cancellation. The second is the definition of a full poly-isomorphism. $\square$

The Boolean countermodel in [Theorem 3.1](#) shows that fullness is essential: for a proper restricted family the conjugate need not be allowed.

6. FIXED-BRANCH RECONSTRUCTION

The actual possible-image set is a union over many choices. To prove existence of one witness, it is not necessary to control all branches simultaneously.

Fix a horizontal representative h , a Kummer input index, the identity representatives for Ind1 and Ind2, and the ordinary Kummer branch for Ind3. The source statements used by this reconstruction are:

- (i) horizontal LGP/lgp links map theta-pilot objects to q-pilot objects;
- (ii) the local upper-semicompatible containers contain the ordinary Kummer image;
- (iii) local fractional ideals globalize to objects of the corresponding global Frobenioids;
- (iv) global log-volume agrees with arithmetic degree after normalization.

Under the structured compatibility hypothesis, the fixed branch produces a global fractional-ideal region W satisfying

$$W \in \mathcal{U}_\Theta, \quad \mu_{\text{proc}}(W) = -|\log q|.$$

The numerical bridge follows.

This is a source-level reconstruction. It is not yet an object generated by a complete Lean encoding of Hodge theaters and Frobenioids.

7. GENERATED POSSIBLE-IMAGE UNIONS

An independent membership field can be removed by defining the theta-pilot region from the concrete outputs.

Let $\mathcal{O}$ be a nonempty type of outputs and let $R_{o,i}$ be the admissible region associated to output o in capsule i . Assume that for each i there is a finite set S_i of rational places such that

$$v_{\mathbb{Q}} \notin S_i \implies R_{o,i}(v_{\mathbb{Q}}) = \mathcal{O}_{i,v_{\mathbb{Q}}}$$

for all outputs o .

Definition 7.1 (Generated theta-pilot union). Define

$$T_i(v_{\mathbb{Q}}) = \bigcup_{o \in \mathcal{O}} R_{o,i}(v_{\mathbb{Q}}).$$

Theorem 7.2 (Admissibility and definitional membership). *Each T_i is a restricted-product admissible region, and*

$$R_{o,i} \subseteq T_i$$

for every o .

Proof. Outside S_i all regions in the union equal the same integral region. Since $\mathcal{O}$ is nonempty, the union equals that integral region there. Hence the support is finite. Membership is the canonical injection into a union. $\square$

This construction was implemented directly against the public `LargeVolumeContainerData.AdmissibleReg` type. It removes the separate field asserting that the native output belongs to a previously supplied theta region.

8. ACTUAL PUBLIC-REGION WITNESS THEOREM

Let

$$X : \text{Iut.Corollary312VariantData}.$$

An actual witness consists of a capsule-indexed family of public admissible regions W_i , containment $W_i \subseteq \Theta_i$, an exact q-volume identity, and monotonicity on the two good region families used in the argument.

Theorem 8.1 (Actual witness closes the public numerical variant). *An actual public-type witness implies*

$$\text{Iut.Corollary312Variant}(X).$$

Proof. The public hull system gives

$$\Theta_i \subseteq H(\Theta_i).$$

Thus $W_i \subseteq H(\Theta_i)$. Using the exact q-volume identity and procession-volume monotonicity gives

$$X.\text{qPilot.lhs} \leq X.\text{rhsData.rhs}.$$

$\square$

This is one of the strongest formal results obtained: the numerical theorem no longer uses an auxiliary region type. The remaining problem is constructing the actual witness from complete IUT source definitions.

9. FINITE-POSITIVE LOG-VOLUME MONOTONICITY

The public volume interface is total on arbitrary regions and therefore does not impose a global monotonicity axiom. For finite positive measure-realized regions, the required theorem is standard.

Theorem 9.1 (Logarithmic measure monotonicity). *Let μ be a measure and let $A \subseteq B$. If*

$$\mu(A) > 0, \quad \mu(B) < \infty,$$

then

$$\log \mu(A) \leq \log \mu(B).$$

Proof. Measure monotonicity gives $\mu(A) \leq \mu(B)$. The real logarithm is increasing on $(0, \infty)$. $\square$

Nonnegative finite weighted sums and finite averages preserve pointwise inequalities. Hence packet and procession monotonicity follow once the actual component log-volumes are represented by finite positive Haar measures.

10. NATIVE Q-PILOT NORMALIZATION

Let

$$D_q = \sum_v n_v [v]$$

be the arithmetic q-divisor, where $n_v = \text{ord}_v(q_v)$ is the integer uniformizer order. Define

$$\log(q) = \deg_{\text{norm}}(D_q).$$

The q-pilot uses a 2ℓ -th root of the Tate parameter, so its realified divisor is

$$D_{\mathbf{q}} = \frac{1}{2\ell} D_q.$$

By linearity,

$$\deg_{\text{norm}}(D_{\mathbf{q}}) = \frac{1}{2\ell} \log(q) = |\log q|.$$

With the signed volume convention,

$$\boxed{\mu_{\text{proc}}(Q_{\text{pilot}}) = -|\log q|}.$$

This closes the scalar calibration of the native object. It does not prove that the same object is an actual member of the theta-pilot possible-image union.

11. RAMIFICATION CORRECTION

Let $L/\mathbb{Q}_p$ have ramification index e and residue degree f . Write

$$p = u\pi^e, \quad q = u_q\pi^n,$$

with units u, u_q . Suppose ρ is the packet-normalized logarithmic response, additive under multiplication, zero on units, and normalized by

$$\rho(p) = -\log p.$$

Then

$$\rho(\pi) = -\frac{1}{e} \log p, \quad \rho(q) = -\frac{n}{e} \log p.$$

Multiplying by the normalized local-degree weight

$$\frac{ef}{[K : \mathbb{Q}]}$$

cancels e and yields

$$-\frac{nf}{[K : \mathbb{Q}]} \log p,$$

which is the normalized arithmetic-divisor contribution.

This proves a normalization fork:

- local-degree weight ef/d must be paired with p-normalized order n/e ;
- integer uniformizer order n must be paired with residue-degree weight f/d .

Using local-degree weight and integer order simultaneously introduces an extra factor e . A ramified quadratic counterexample was formalized.

12. PACKET MARGINALIZATION AND PROCESSION AVERAGING

Let S be a finite label set, let V_p be the finite fiber over p , and let $w : V_p \rightarrow \mathbb{R}$ satisfy

$$\sum_{v \in V_p} w(v) = 1.$$

A packet component is a function $c : S \rightarrow V_p$ with product weight

$$W(c) = \prod_{j \in S} w(c(j)).$$

Theorem 12.1 (Product-weight marginalization). *For every distinguished label $j_0 \in S$ and every $f : V_p \rightarrow \mathbb{R}$,*

$$\sum_{c:S \rightarrow V_p} W(c)f(c(j_0)) = \sum_{v \in V_p} w(v)f(v).$$

Proof. Split c into its j_0 coordinate and the remaining coordinates. For a fixed value of the distinguished coordinate, the sum of the weights over the remaining coordinates is the product of copies of $\sum_v w(v) = 1$. $\square$

More generally, any barycentric allocation across labels whose coefficients sum to one has the same normalized reading. A naive diagonal action at every label is impossible: it multiplies a nonzero response by the capsule cardinality, which varies along the procession.

If every capsule has the same calibrated global q-reading, its procession average is that same value. This removes procession averaging as an independent hypothesis.

13. HOLOMORPHIC HULL REDUCTIONS

Two independent reductions were obtained.

First, if a generated union lies inside a log-shell whose closure is compact, then the closure of the generated union is compact. This proves the relative compactness portion of hull admissibility.

Second, suppose a family of regions is indexed by exponents in an antitone chain. If the exponent family has an attained lower bound m , then the region at exponent m is the least member of the chain containing the union. This proves the order-theoretic portion of a least-hull construction.

The remaining local-field task is to instantiate the chain by the actual nonarchimedean lattices and the archimedean closed balls or polydiscs in the IUT packet summands.

14. THE IUT IV DOWNSTREAM RECONSTRUCTION

Once the corrected Corollary 3.12 estimate is available, the final scalar algebra is elementary. A typical coefficient has the form

$$C_\Theta = \lambda \left(A - \frac{1}{6} \log q \right) - 1, \quad \lambda > 0.$$

Then $C_\Theta \geq -1$ implies

$$\frac{1}{6} \log q \leq A.$$

The subsequent ε -absorption inequalities have also been formalized.

The geometric part has three standard components.

14.1. The Legendre height comparison. For

$$E_\lambda : y^2 = x(x-1)(x-\lambda),$$

the j -invariant is

$$j(\lambda) = 2^8 \frac{(1-\lambda+\lambda^2)^3}{\lambda^2(1-\lambda)^2}.$$

Its pole divisor is $2[0] + 2[1] + 2[\infty]$. Hence

$$h_\infty(j(\lambda)) = 6h_{\omega_{\mathbb{P}^1}(0+1+\infty)}(\lambda) + O(1).$$

Tate uniformization identifies $\log q$ with the pole-height of j up to bounded discrepancy on the relevant compactly bounded family, yielding

$$\frac{1}{6} \log q = h_{\omega_{\mathbb{P}^1}(0+1+\infty)} + O(1).$$

14.2. Tripod different and conductor. For a coprime triple $a + b = c$ and $\lambda = a/c$,

$$h(\lambda) = \log \max\{a, b, c\},$$

the relative different over $\mathbb{Q}$ is zero, and

$$\log \text{Cond}_{\{0,1,\infty\}}(\lambda) = \sum_{p|abc} \log p = \log \text{rad}(abc).$$

Thus the tripod Vojta inequality specializes exactly to the logarithmic abc inequality.

14.3. Exceptional sets and admissible primes. A finite exceptional set can be absorbed into the bounded-discrepancy constant. The elementary prime-selection step — choosing a prime above a bound and outside a finite set — is also formalized. The non-elementary part is proving that the Galois-image, Tate-order, and initial-theta conditions fail only on an admissible exceptional family; that remains part of the actual IUT IV source theorem.

15. TRIPOD STATEMENT II IMPLIES ABC

Let T be an actual geometric tripod height theory. Suppose there is an encoding of abc triples into points of T and constants A_h, A_d, A_c such that

$$\log \max\{a, b, c\} \leq h_T(P) + A_h,$$

$$\log \text{Diff}_T(P) \leq A_d,$$

$$\log \text{Cond}_T(P) \leq \log \text{rad}(abc) + A_c.$$

If Statement II gives, uniformly in P ,

$$h_T(P) \leq (1 + \varepsilon)(\log \text{Diff}_T(P) + \log \text{Cond}_T(P)) + B_\varepsilon,$$

then the logarithmic abc conjecture follows after absorbing the comparison constants. This implication is formalized without storing abc as a field.

16. THE QUANTIFIER CORRECTION

A universal Diophantine theorem cannot follow from one isolated initial Theta datum. The correct source is a dependent family

$$x \mapsto (D_x, Q_x, S_x)$$

indexed by all arithmetic inputs to which the theorem applies.

A finite Boolean countermodel formalizes the logical point:

$$\exists x \text{ Certificate}(x) \not\Rightarrow \forall x \text{ Certificate}(x).$$

Thus the first two construction goals must have the form

$$\forall x : \text{Input}, \quad D_x : \text{InitialThetaData},$$

$$\forall x : \text{Input}, \quad S_x : \text{GeneratedNativeSource}(D_x, Q_x).$$

17. CIRCULARITY OF THE FORMER DOWNSTREAM CERTIFICATE

An earlier certificate used a field

$$(\forall x, \text{Cor312}(x)) \longrightarrow \text{ABCConjecture}.$$

For a fixed source family F , the type of such downstream objects is inhabited if and only if the target proposition is already true:

$$\boxed{\text{Nonempty}(\text{UniformDownstream}(F, T)) \iff T.}$$

Consequently the former three-closure certificate was inhabited precisely when an upstream family existed and abc was already true. This circularity was proved in Lean.

18. THE NON-CIRCULAR FINAL CERTIFICATE

The circular field is replaced by explicit intermediate data:

- an encoding of *abc* points into the input family;
- functions $\log Q$, $\log \text{Diff}$, and $\log \text{Cond}$;
- uniform height, different, and conductor comparison constants;
- for each $\varepsilon > 0$, one constant C_ε such that Corollary 3.12 for the encoded input implies the uniform q -height estimate.

This structure is called a non-circular IUT IV bridge. Lean proves that a pointwise actual IUT III source family together with this bridge implies the standard logarithmic *abc* conjecture.

19. THE FOUR-STAGE PROGRAMME

The final honest formal structure has three construction stages and one consequence:

- (1) an actual initial Theta-data family;
- (2) an actual IUT III multiradial source family;
- (3) an actual non-circular uniform IUT IV geometric bridge;
- (4) the theorem *abc*.

The Lean project defines a structure `FourStageProgram` containing the first three stages and proves

$$\boxed{\text{FourStageProgram} \implies \text{ABCConjecture.}}$$

Equivalently,

$$\text{Nonempty}(\text{FourStageProgram}) \implies \text{ABCConjecture.}$$

No inhabitant of `FourStageProgram` is constructed.

20. WHY THE FOUR MISSING CONSTRUCTIONS WERE NOT SOLVED HERE

The remaining problem is not a shortage of wrapper definitions. It is the absence of proofs inhabiting the fields.

20.1. Actual initial Theta-data family. The public structure contains a number field, an elliptic curve, an admissible prime, a genuine $\text{mod-}\ell$ representation, a torsion field, orbicurves, coverings, fundamental groups, core relations, cusps, Tate parameters, and local conditions. Defining a record with these field names does not construct these objects or prove their compatibility.

20.2. Actual IUT III source family. The public Corollary 3.12 project takes the theta region, packet presentation, log-volume, and hull system as interface data. It does not define a complete Hodge-theater/Frobenioid/log-link/Kummer/multiradial source whose output inhabits those interfaces. The generated-union construction proves what to do with concrete outputs; it does not manufacture the outputs.

20.3. Actual IUT IV uniform bridge. The non-circular bridge requires one uniform constant for all arithmetic points and all the actual height, different, and conductor comparisons. It also requires the source theorems controlling exceptional sets, admissible primes, and the actual local/global packet estimates. Packaging these statements as fields does not prove them.

20.4. Parameter-free *abc* theorem. The final theorem is a consequence of the preceding stages, not an additional independent construction. A declaration

`theorem abc_conjecture : ABCConjecture`

without a term inhabiting the first three stages would hide an unproved axiom.

21. LEDGER OF RESULTS

Result	Status	Comment
Standard logarithmic <i>abc</i> statement	Formalized	Exact target proposition
Weak common-source sufficiency	Refuted	Finite Boolean counter-model
Composite poly-morphism extraction	Proved	Elementary set-valued composition
Full-poly Kummer conjugation	Proved	Requires fullness at the correct level
Ordinary generated membership	Proved	Definitional in generated choice type
Actual public witness $\Rightarrow$ public Cor312 variant	Proved	Uses public region/hull types
Generated theta union admissible	Proved conditionally	Requires common finite support
Finite-positive log-volume monotonicity	Proved	Standard measure theory
Native q-pilot degree calibration	Proved abstractly	Uses linear arithmetic degree
Ramification correction	Proved	Local response is $-(n/e) \log p$
Packet marginalization	Proved	Finite product-measure identity
Procession averaging	Proved	Constant capsule reading
Finite exceptional-set absorption	Proved	Does not prove Galois-finiteness
Elementary prime avoidance	Proved	Does not prove large Galois image
IUT IV scalar algebra	Proved	Does not prove actual packet estimates
Tripod Statement II $\Rightarrow abc$	Proved conditionally	Requires actual comparison package
Pointwise quantifier correction	Proved	One certificate is insufficient
Circularity of old downstream certificate	Proved	Old certificate already contained <i>abc</i>
Non-circular bridge $\Rightarrow abc$	Proved	Honest final logical closure
Actual initial Theta-data family	Open	No source inhabitant constructed
Actual IUT III source family	Open	Complete source types absent
Actual uniform IUT IV bridge	Open	Uniform geometric theorem absent
Parameter-free <i>abc</i> theorem	Open	Equivalent to inhabiting prior stages

22. LEAN PACKAGE ARCHITECTURE

The accompanying project contains thirty-seven Lean modules. Its main groups are:

Public Corollary 3.12 adapter:

`ActualPilotWitness.lean`, `GeneratedSource.lean`.

Quantifier and circularity audit:

`QuantifierCorrectClosure.lean`, `CircularityAudit.lean`, `NonCircularDownstream.lean`.

Full-poly and countermodel core:

FullPolyCore.lean, WeakCompatibilityCountermodel.lean.

Q-pilot normalization:

QPilotNormalizationAudit.lean, QPilotNormalizationFork.lean, RootQPilotDivisor.lean, RamificationCorrectedQPilot.lean.

Packets, volumes, and hulls:

FinitePositiveLogVolumeMonotonicity.lean, ProductWeightMarginalization.lean, PrimePowerQPilot, NativeQPilotCalibration.lean, FiniteExponentHull.lean.

Downstream algebra:

FiniteExceptionalSet.lean, AdmissiblePrimeSelection.lean, Cor312CoefficientAlgebra.lean, IUTIVAbsorption.lean.

Final honest closure:

FourOpenConstructions.lean, AxiomAudit.lean.

The source contains no explicit `sorry`, `admit`, or user-declared `axiom`. The current runtime did not contain Lean, so the v4.0 archive received static auditing but not a fresh local kernel build.

23. CONCLUSION

The research programme has reached a sharp boundary. The categorical, measure-theoretic, normalization, averaging, scalar-algebra, quantifier, and final-transfer components are no longer vague. They have been stated as precise theorems, many of them formalized.

What remains is substantially deeper: actual construction of the source objects that the public Lean interfaces deliberately leave abstract. No amount of renaming certificates can replace the existence theorems for those objects. Accordingly, the present work is a comprehensive reconstruction and formalization audit, but it does not claim an unconditional solution of the *abc* conjecture.

APPENDIX A. EXACT NON-CIRCULAR THEOREM

The formal theorem has the schematic shape

pointwise actual IUT III source family
 + uniform q-height estimate from Corollary 3.12
 + height/different/conductor comparisons
 $\implies ABCConjecture$.

The exact remaining proposition is

Nonempty(FourStageProgram).

APPENDIX B. BIBLIOGRAPHIC SOURCES

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